3. Co-dominance and certain homozygous genotypes can give unusual ratios in the resulting offspring. **Image 3.1** shows some phenotypes of Camellia plants.

Image 3.1

Variegated leaves (green and white)

Green leaves

Red flower

Red and white flower

Horticulturalists crossed plants which had red flowers and variegated leaves with plants which had red and white flowers and variegated leaves.

(a) (i) Using the symbols below show this cross and the resulting offspring by:

- completing the parental genotypes and gametes;
- drawing a genetic diagram. [2]

RR red flower
RW red and white flower
WW white flower

GG green leaves
GA variegated leaves
AA white leaves

Parental phenotype	red flower, variegated leaves	Parental phenotype	red and white flower variegated leaves
Parental genotype	....	Parental genotype	....
Gametes	....	Gametes	....

Genetic Diagram

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(ii) Use the genetic diagram drawn in (a)(i) to complete **Table 3.2**. [3]

Table 3.2

Genotypes		....	....	....	....	....	....
Expected phenotypes	Flower colour	....	....	....	....	....	....
	Leaf type	....	....	....	....	....	....
Expected ratio		....	....	....	....	....	....

(b) The resulting 420 seeds from the above cross were planted and the phenotypes of all the **mature** plants were counted and the following results observed.

56 red flowers, green leaves
102 red flowers, variegated leaves
49 red and white flowers, green leaves
110 red and white flowers, variegated leaves

The following observation was made:

'Of the 420 seeds produced in the above cross, all the seeds germinated but 103 did not grow and so a different phenotypic ratio was observed than expected.'

Explain this observation. Your answer should include reference to:
- germination;
- plant leaf cell structure;
- observed and expected phenotypic ratios. [5]

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(c) The production of seeds requires sexual reproduction which involves meiosis. One way for a plant grower to produce plants with the same phenotype as the parent plant is to take cuttings. Using your knowledge of cell division explain why the same phenotype would be retained. [3]

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